

Year 1

Subject – Design

Unit-1

Summative



Image source:

https://gict.co.in/wp-content/uploads/2024/07/Cover_Interaction-design.webp

Designing for Interaction

Design Situation:

Young learners use many digital systems every day such as games, apps and learning platforms. These systems communicate with users through inputs, outputs and interactive responses.

In this unit, students will explore how simple **block-based coding** can be used to **design interactive digital solutions**. Through the **design cycle**, they will investigate user needs, develop ideas, create a simple interactive project and evaluate its effectiveness.

Students will create an **interactive project** using Pictoblox that demonstrates communication between a user and a digital system.

MYP 1 Unit-1 Elements

- **Specified Concepts:** Communication, Function, Systems, Innovation
- **Global Context:** Scientific and Technical Innovation (Products, Processes and Solutions)
- **Statement of Inquiry:** Understanding how systems and functions enable communication helps individuals design innovative interactive solutions that respond to user needs.
- **ATL Skills:** Thinking Skills, Self-management skills, Research skills, Communication Skills
- **IB Learner Profile:** Thinker

Command Terms

Command Term	Definition
Analyze	Break down in order to bring out the essential elements or structure. (To identify parts and relationships and to interpret information to reach conclusions.)
Create	To evolve from one's own thought or imagination, as a work or an invention.
Demonstrate	Make clear by reasoning or evidence, illustrating with examples or practical application.
Describe	Give a detailed account or picture of a situation, event, pattern or process.
Develop	To improve incrementally, elaborate or expand in detail. Evolve to a more advanced or effective state.
Evaluate	Make an appraisal by weighing up the strengths and limitations.
Explain	Give a detailed account including reasons or causes. (See also "Justify".)
Identify	Provide an answer from a number of possibilities. Recognize and state briefly a distinguishing fact or feature.
Justify	Give valid reasons or evidence to support an answer or conclusion. (See also "Explain".)
List	Give a sequence of brief answers with no explanation.
Outline	Give a brief account or summary.
Present	Offer for display, observation, examination or consideration.
State	Give a specific name, value or other brief answer without explanation or calculation.

A Description of Your Grasp Summative Assessment

Designing the product

G The goal is to design and create a simple interactive digital solution using Pictoblox that demonstrates how a digital system communicates with users through inputs, outputs and programmed responses.

R Your role will be a **young digital designer** responsible for creating an interactive project that engages users and demonstrates effective communication through coding.

A Your audience includes your classmates, teachers and young learners who will interact with and test your digital solution.

S You are in a situation that many digital systems require interaction between users and technology. Users provide information and expect meaningful responses. You have been challenged to design an interactive project that allows users to interact with a digital system using inputs, outputs and simple programmed logic. Your project should be easy to use, engaging and functional.

P The product is to design and create a guided interactive project using Pictoblox. Possible examples are:

- Guess the Number Game
- Simple Quiz
- Greeting Program
- Choice-based Interaction

S This stands for standards and criteria for success. Your work will be assessed using **all four MYP Design Criteria (A, B, C and D)**, focusing on research and analysis, idea development, creation and evaluation of your final coded solution.

Inquiry Questions

Factual: (Students need to know)

- What is a program?
- What are input and output?
- What are algorithms and flowcharts?

Conceptual: (Connect to specified concepts)

- How do systems and functions work in interactive programs?
- How does interaction improve communication?

Debatable: (To Encourage Discussion)

- Technology always improves communication

Contextual Lenses

Individual Lens

How do I interact with digital systems in my daily life?

Local Lens

How do people in my school or community use interactive applications?

Global Lens

How do interactive technologies impact communication worldwide?

Criteria D: Evaluating

Before completing your project, you must test your solution, evaluate its success, identify possible improvements and reflect on its impact on users.

Strand i: Outline simple, relevant testing methods, which generate data, to measure the success of the solution

Activity: Testing the Solution

Thinking Skills – Critical Thinking – Test generalizations and conclusions

To find out if your solution works successfully, you must test it with users.

Task: Describe how you tested your solution.

Tester	Feedback
Friend	
Parent	
Classmate	

Test Conducted	Result
User entered input	
Program provided output	
Conditional statement worked correctly	
Program responded as expected	

Strand ii: Outline the success of the solution against the design specification

Activity: Evaluating Against Success Criteria

Self-management Skills – Reflection – Consider content (What did I learn? What don't I yet understand?)

Review the success criteria you created in Criterion B.

Success Criteria	Achieved? (Yes/No)	Evidence/Explanation
1		
2		
3		
4		
5		

Strand iii: Outline how the solution could be improved

Activity: Improvements

Thinking Skills – Creativity and Innovation – Design improvements to existing machines, media and technologies

If you had more time, what improvements would you make?

Improvement	Why would it help?

Strand iv: Outline the impact of the solution on the client / target audience

Activity: Impact on Users

Communication Skills – Give and receive meaningful feedback

Who is the target audience for your solution?

How did your solution help, engage or communicate with users?

What feedback did users give about your solution?

Reflection

What did you learn from completing this project?

What would you do differently next time?

Bibliography: (MLA9 format)

The bibliography lists all the sources you used during your research. MLA 9 is the referencing style commonly used in the IB Middle Years Programme.

Examples:

Book

- Resnick, Mitchel. *Lifelong Kindergarten: Cultivating Creativity through Projects, Passion, Peers, and Play*. MIT Press, 2017.

Website (PictoBlox)

- STEMpedia. *PictoBlox*. STEMpedia, <https://thetempedia.com/product/pictoblox/>. Accessed 27 June 2026.

Website (Scratch Ideas and Coding Concepts)

- Scratch Foundation. *Scratch*. <https://scratch.mit.edu/>. Accessed 27 June 2026.

Educational Website

- Code.org. *Learn Computer Science*. <https://code.org/>. Accessed 27 June 2026.

Online Tutorial

- STEMpedia. *PictoBlox Documentation*. <https://ai.thetempedia.com/docs/pictoblox/>. Accessed 27 June 2026.

Video

- STEMpedia. *Getting Started with PictoBlox*. YouTube, <https://www.youtube.com/@STEMpedia>. Accessed 27 June 2026.

Personal Interview

- Personal interview with your Design teacher about interactive digital systems, June 2026.

Design Summative Assessment Rubric, Year1

Student Name: _____

Criteria D: Evaluating

Achievement Level: /8

At the end of this unit 1, year 1 students should be able to:

- i. outline simple, relevant testing methods, which generate data, to measure the success of the solution.
- ii. outline the success of the solution against the design specification.
- iii. outline how the solution could be improved.
- iv. outline the impact of the solution on the client/target audience.

Achievement level	Achievement level descriptor	Task specific clarifications
0	The student does not reach a standard described by any of the descriptors below.	The student does not demonstrate evidence of testing, evaluation, improvement, or impact.
1–2	The student: i. outlines one simple testing method ii. states whether the solution is successful or not iii. states one improvement iv. states one impact of the solution	Your response shows that you have: • outlined one simple testing method such as self-testing or asking one user to try the solution. • made a basic statement about whether the solution worked. • suggested one simple improvement. • stated one way the solution affected the user or audience.
3–4	The student: i. outlines a few simple, relevant testing methods ii. outlines the success of the solution against some design specifications iii. outlines a few improvements iv. outlines the impact of the solution on the client/target audience	Your response shows that you have: • used more than one testing method and recorded basic results or feedback. • compared your solution against some success criteria from Criterion B. • suggested a few realistic improvements. • described how the solution affected users using some feedback.
5–6	The student: i. outlines relevant testing methods that generate data ii. outlines the success of the solution against the design specification iii. outlines how the solution could be improved iv. outlines the impact of the solution on the client/target audience	Your response shows that you have: • used several relevant testing methods and collected clear results or user feedback. • evaluated your interactive solution against most of the success criteria. • suggested realistic improvements based on testing results. • clearly explained how the solution helped, engaged or communicated with the target audience.

7-8	The student: - outlines detailed, relevant testing methods that generate data - outlines the success of the solution against the design specification - outlines how the solution could be improved - outlines the impact of the solution on the client/target audience	Your response shows that you have: - planned and carried out multiple relevant tests and collected meaningful user feedback and data. - clearly evaluated the success of your solution against all success criteria developed in Criterion B. - suggested well-reasoned and realistic improvements supported by evidence from testing. - clearly explained the impact of the solution on the target audience, supported by user feedback and testing results.
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Comments:
